

Exclusive Encapsulation Adhesive in a Neutral-Plane Model for Ultrahigh Mechanical Stability of Flexible Perovskite Solar Cells

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The functional layers of flexible perovskite solar cells (FPSCs) are subjected to significant stress during bending, causing structural failure and declining power conversion efficiency. To alleviate stress, a neutral plane (NP) is introduced by positioning the functional layers at the center of the device, with top metal electrode encapsulated by a protective layer. However, adhesive detachment remains a critical issue during multiple bending cycles, and the underlying mechanism remains unclear. In this study, a systematic analysis of the detachment behavior in NP-FPSC reveals that commercial adhesives with high Young's modulus and low adhesion strength struggle to withstand interlayer shear stress during bending, which triggers detachment between adhesives and electrodes. To address this issue, a crosslinkable polymer acrylated isoprene rubber (AIR) is designed with long linear polyisoprene main chain and acrylate side chains, providing high flexibility and reduced chain segment movement. AIR can be crosslinked under UV irradiation to form a stable network with ultralow Young's modulus and high adhesion strength, ensuring a strong bond between the protective layer and FPSCs, constructing stable NP-FPSCs. The resultant NP-FPSCs demonstrate excellent mechanical stability, retaining 92.8% of their initial efficiency after 50 000 bending cycles at a radius of 4 mm, meeting the IEC 62715-6-3 standards.

energy systems owing to their exceptional optoelectronic properties, lightweight design, and high specific power density.^[1] Bending resistance is essential in such applications to maintain efficiency and device stability under continuous bending and other mechanical stresses.^[2] However, the current mechanical stability of FPSCs still cannot meet these application demands.

State-of-the-art FPSCs are typically constructed by a layer-by-layer configuration on a plastic substrate, with the perovskite layer sandwiched between electron and hole transport layers.^[3] Because the plastic substrate is significantly thicker than the device, these layers experience the greatest stress during bending, according to the neutral plane (NP) mechanics model.^[4] When applied stress exceeds the elastic deformation tolerance of materials, structural failure initially occurs in the polycrystalline perovskite film and eventually propagates to the interlayers, leading to optoelectronic performance decline.^[5] Efforts have been devoted to improving bending fatigue resistance by reducing the Young's

modulus of bulk perovskite films and enhancing interlayer toughness.^[5b,6] However, FPSCs are still compromised under extreme conditions (e.g., smaller bending radius and more frequent bending cycles), due to the brittle nature of the indium tin

1. Introduction

Flexible perovskite solar cells (FPSCs) have great potential for wearable devices, building-integrated photovoltaics, and mobile

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DOI: 10.1002/adma.202501776

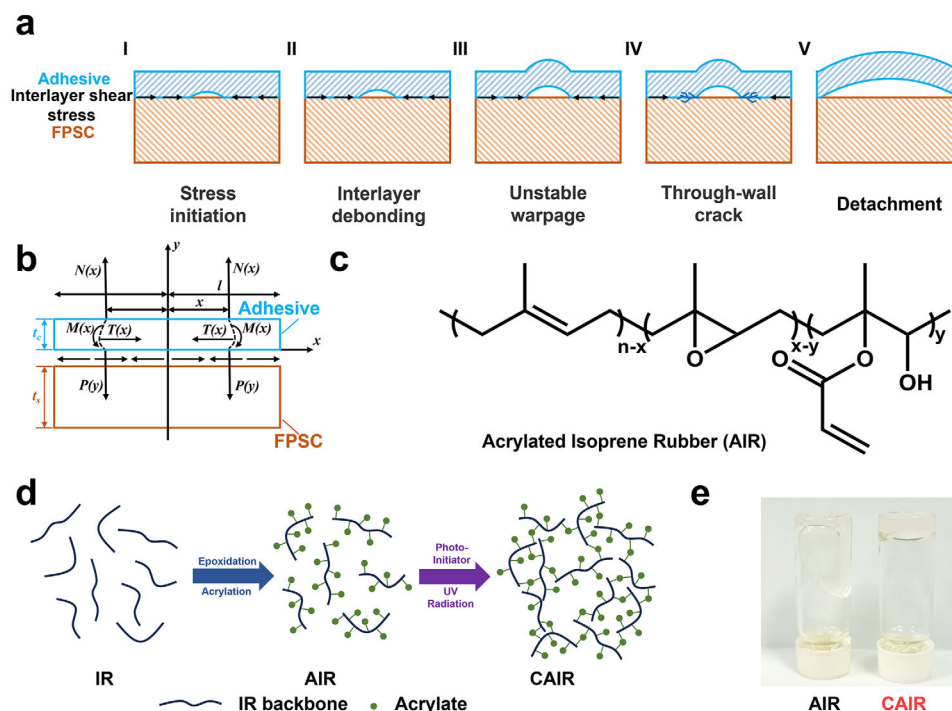


Figure 1. a) Failure process of combined interlayer. b) Failure stress model of combined interlayer. c) Molecular structure of AIR. d) The schematic diagram of synthetic route and crosslinking process of AIR. e) Photographs of AIR and CAIR products.

oxide (ITO) electrode and perovskite, those with small Poisson's ratio and large Young's modulus.^[7] Therefore, minimizing mechanical stress on functional layers under extreme conditions is essential for achieving mechanically robust FPSCs.

According to structural mechanics theory, positioning functional layers of FPSCs at the NP eliminates force during bending.^[3c,8] Lee et al. constructed NP-FPSC by depositing a parylene protective film onto the FPSC using chemical vapor deposition (CVD), shifting the NP to the perovskite film. This reduced tensile stress on the perovskite layer, even at a bending radius of 0.5 mm.^[1a] However, CVD increases manufacturing cost. Yang et al. used commercial hot-melt adhesive film, surlyn, to affix a protective polyethylene terephthalate (PET) film onto FPSCs through hot-pressing, creating NP-FPSCs.^[9] This shifts functional layers toward the NP, relieving stress during bending and achieving an eightfold improvement in bending lifetime compared to fresh devices. However, the adhesive layer may still detach from the device after long-term bending,^[10] disrupting the NP structure and subjecting all functional layers to high stress once again. Most importantly, the underlying mechanism of this detachment remains unclear. Therefore, it is crucial to investigate this mechanism and develop advanced adhesives for constructing durable NP-FPSCs.

In this study, we investigated the detachment behavior at the device/adhesive interlayer of an NP-FPSC under mechanical stress. We found that commercial adhesives with high Young's modulus and low adhesion strength struggle to withstand significant shear stress during repeated bending, leading to detachment from electrodes. To address this issue, we designed a crosslinkable polymer acrylated isoprene rubber (AIR) with a long linear polyisoprene backbone and acrylate side groups. This structure

provides high flexibility and reduced chain segment movement. AIR effectively crosslinks under UV irradiation to form a compact, stable network with ultralow Young's modulus and high adhesion strength, ensuring a strong bond between the protective layer and FPSCs. As a result, stable NP-FPSCs are constructed, exhibiting excellent mechanical stability and retaining 92.8% of their initial efficiency after 50 000 bending cycles at a radius of 4 mm, meeting the IEC 62715-6-3 standards.

2. Results and Discussion

The NP-FPSCs were constructed by affixing a PET protective layer onto the flexible device using an adhesive (Figure S1, Supporting Information). Here, the F-PSCs is treated as a single layer because its functional layers ($\approx 1 \mu\text{m}$) are negligible in thickness compared to the PET substrate ($\approx 125 \mu\text{m}$).^[2a,11] Considering that the adhesive layer ($\approx 30 \mu\text{m}$, measured by profilometer) is considerably thinner than both the PET protective layer ($\approx 125 \mu\text{m}$) and the entire flexible device ($\approx 126 \mu\text{m}$), we investigated and modeled the detachment behavior of NP-FPSCs according to the classical coating-substrate interlayer model.^[12] The adhesive layer is labelled as the coating, while the whole FPSCs is regarded as the substrate. Figure 1a illustrates the adhesive detachment process, occurring in five stages: I) differences in Young's modulus, Poisson's ratio, and thickness between the adhesive and the whole FPSCs subject their interlayer to significant shear stress when bent. II) Once this stress exceeds a critical threshold (interlayer adhesion strength), the adhesive undergoes interlayer debonding from the whole FPSCs, forming an interlayer crack.^[13] III) Under continuous shear stress, the crack propagates until the adhesive detaches from the whole FPSCs, causing warpage.^[14]

IV) The interlayer shear strain transfers to bulk strain in the adhesive layer, resulting in elastic deformation and through-wall crack formation.^[15] V) Finally, the adhesive layer peels off from the whole FPSCs, completing detachment.^[16] Thus, detachment is determined by interlayer shear stress and adhesion strength.

Figure 1b illustrates the geometry and stress conditions for quantitatively analyzing the interlayer shear stress when the interlayer lies between the adhesive with a thickness of t_c and the electrode with a thickness of t_s . During bending, various interaction forces are generated, e.g., $M(x)$ —bending moment applied to the model; $T(x)$ —interlayer shear force between adhesive and electrode; $N(x)$ —normal pressure between adhesive and electrode; $P(y)$ —peel force between adhesive and electrode. Based on these forces and an interlayer failure model, the interlayer shear stress can be evaluated using the following equation (the detailed derivation process is provided in Note S1 (Supporting Information^[17]):

$$\tau(x) = E_c t_c \Delta t \sqrt{\frac{\lambda}{k}} \quad (1)$$

where $\tau(x)$ is the interlayer shear stress between the adhesive and electrode; E_c is the Young's modulus of the adhesive; t_c is the adhesive thickness; Δt is the thickness difference between the adhesive and electrode; $\lambda = \frac{1}{E_s t_s} + \frac{1}{E_c t_c}$; E_s is the Young's modulus of the electrode material; t_s is the electrode thickness; $k = \frac{2(1+\nu_c)t_c}{3E_c} + \frac{2(1+\nu_s)t_s}{3E_s}$; ν_c is the Poisson's ratio of the adhesive; and ν_s is the Poisson's ratio of the electrode.

Here, λ can be approximated as $\frac{1}{E_c t_c}$, and k can be approximated as $\frac{2(1+\nu_c)t_c}{3E_c}$ because of the condition $E_c \ll E_s$. Thus, Equation (1) can be expressed as follows:

$$\tau(x) \approx \frac{3E_c \Delta t}{2(1+\nu_c)} \quad (2)$$

According to Equation (2), interlayer shear stress can be reduced by lowering the adhesive's Young's modulus. A high adhesion strength of interlayer is also required to resist shear stress induced detachment. Crosslinked isoprene rubber (IR) is an excellent adhesive with a low Young's modulus, widely used in semiconductors, tires, metals, and biomedical science.^[1c,18] Its crosslinking methods usually need small molecule crosslinking agents or polar solvents, which can negatively impact the PCE of perovskite solar cells, limiting their application in NP-FPSCs.^[19] We synthesized acrylated isoprene rubber (AIR, molecular structure shown in Figure 1c) by introducing acrylate-functionalized side groups into IR, avoiding the need for additional crosslinking agents and enabling photocuring at room temperature. The introduction of epoxy groups in AIR increases the rigidity of polyisoprene molecular chains and limits the mobility of molecular segments. Additionally, epoxy groups have high polarity and reactivity, which enhance interactions with surrounding molecular chains. These interactions further restrict molecular motion and reduce the chain segment movement between them.^[20] Furthermore, the introduction and crosslinking of acrylic ester side chains create a three-dimensional network structure, further confining polymer segment mobility and occupying intermolec-

ular space, ultimately reducing chain segment movement.^[21] These structural features significantly contribute to the high adhesion strength.^[22] AIR was synthesized through IR epoxidation, followed by ring-opening addition of epoxy units using acrylic acid (Figure 1d; details in "Material Synthesis" section of Supporting Information and Figure S2, Supporting Information).^[23] The successful introduction of acrylic ester groups into AIR was confirmed by the characteristic peaks of vinyl bonds in the proton nuclear magnetic resonance spectrum (6.5, 6.1, and 5.9 ppm) and a carbonyl peak at 1726 cm⁻¹ in the Fourier transform infrared spectrum (Figures S3 and S4, Supporting Information). Moreover, the molar content of acrylate group is 1.61%, which can be calculated by the integral of correlated characteristic peak ($\frac{\text{peak u}}{\text{peak u} + \text{peak o} + \text{peak k}} \times 100\%$) (Figure S3, Supporting Information). The obtained AIR was a colorless, transparent, viscous liquid at room temperature (Figure 1e, left). Subsequently, crosslinked AIR (CAIR) was prepared by photocuring AIR with a trace amount of photoinitiator, diphenyl (2,4,6-trimethylbenzoyl) phosphine oxide (TPO), under 365 nm UV light for 5 min at room temperature. The reaction equation for the crosslinking process of AIR is presented in Figure S5 (Supporting Information), resulting in a solid with adhesion ability (Figure 1e, right). The gel permeation chromatography (GPC) analysis determined the number-average molecular weight (M_n) and weight-average molecular weight (M_w) to be 24100 g/mol and 28000 g/mol, respectively (Figure S6, Supporting Information).

The adhesion strength of CAIR was evaluated using a single-lap tensile shear test. As illustrated in Figure 2a, two glass sheets bonded with CAIR were stretched in opposite directions until detachment. The load at this point was defined as the fracture load (Figure 2c), and the adhesion strength was calculated by dividing the fracture load by the cross-sectional area.^[24] Since IR is not a viable adhesive, we selected two commercially available adhesives for comparison—DELO (KATIBOND LP655), widely used as a photocurable adhesive in the photovoltaic field, and TB (ThreeBond 3178), commonly applied in flexible electronics. The adhesion strength of both adhesives was also evaluated. The adhesion strength of CAIR was significantly higher (3.40 MPa) than those of DELO (0.24 MPa) and TB (0.03 MPa) (Figure 2b; Table S1, Supporting Information) owing to the reduced chain segment movement of CAIR. Their Young's modulus was evaluated using uniaxial tensile test (Figure 2d; Figure S7, Supporting Information). The stress-strain curves of the adhesives were recorded at ambient temperature with a crosshead speed of 2.5 mm min⁻¹ (Figure 2e).^[12,25] The Young's modulus of CAIR, DELO and TB, calculated from the slopes of the stress-strain curve, were 0.15, 8500, and 1.91 MPa, respectively (Figure 2f–h; Table S2, Supporting Information). The high Young's modulus of DELO could be attributed to the restricted segment movement caused by its high crosslinking density.

The interlayer shear stress (τ) of different adhesives was calculated using Equation (2) and summarized in Tables S3 and S4 (Supporting Information). Detachment behavior is governed by the relationship between τ and the stress limit (Σ_a); that is, once τ exceeds Σ_a (e.g., $\tau > \Sigma_a$), the adhesive starts to detach. Σ_a is calculated by multiplying the adhesion strength by the adhesive thickness. The calculated Σ_a for DELO (7.2 N m⁻¹) and TB (0.9 N m⁻¹) are significantly smaller than their respective τ values (9.75

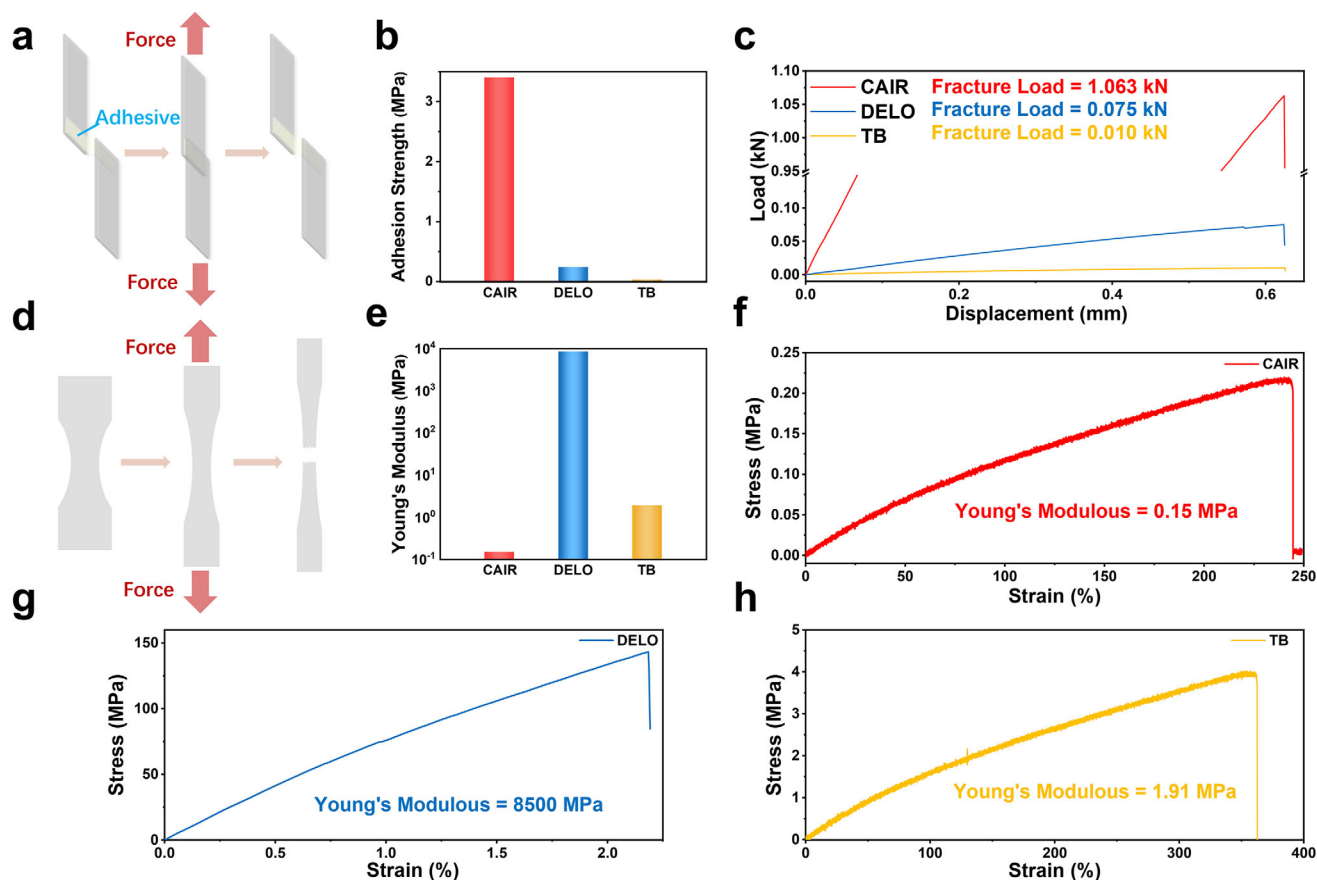


Figure 2. a) Schematic diagrams of the bond strength test. b) The adhesion strength of CAIR, DELO and TB. c) Load-displacement curve of CAIR, DELO and TB. d) Schematic diagrams of the tensile test. e) The Young's modulus of CAIR, DELO and TB. Stress-strain curve of f) CAIR, g) DELO and h) TB.

$\times 10^5 \text{ N m}^{-1}$ and $1.82 \times 10^2 \text{ N m}^{-1}$, respectively), indicating a high likelihood of detachment at their adhesive/electrode interlayers. In contrast, the calculated Σ_a ($1.02 \times 10^2 \text{ N m}^{-1}$) of CAIR is considerably higher than its τ ($1.44 \times 10^1 \text{ N m}^{-1}$), suggesting that interlayer detachment is unlikely to occur. It is noteworthy that CAIR exhibits superelasticity (Figure S8, Supporting Information) and negligible viscoelastic behavior (Figure S9, Supporting Information). Consequently, the Young's modulus of CAIR remains constant during deformation (Figures S9 and S10, Supporting Information). These findings demonstrate that CAIR can withstand interlayer shear stress and prevent detachment during bending. Furthermore, the critical energy release rate (G_c) of the adhesive/electrode interface was investigated using the DCB delamination method (Figure S11 and Note S2, Supporting Information) to assess interfacial adhesion energy. As shown in Figure S12 (Supporting Information), the CAIR/electrode interface exhibited the highest G_c value (43.2 J m^{-2}), significantly higher than that of the DELO/electrode (4.0 J m^{-2}) and TB/electrode (18.9 J m^{-2}) interfaces, thereby effectively preventing interlayer detachment.

We then constructed the NP-FPSCs by attaching a protective PET layer to the Ag electrode using various adhesives with a structure of PET-ITO/SnO₂/Perovskite/Spiro-OMeTAD/Ag/Adhesive/PET (Figures S13 and S14, Supporting Information). Details of the fabrication process are provided in

Note S3 and Figure S15 (Supporting Information). The thickness of the PET protective layer is the same as that of the bottom PET substrate ($125 \mu\text{m}$) to position the entire flexible device near the NP. This design minimizes stress and strain on the functional layers of FPSCs, irrespective of the bending degree (Note S4, Supporting Information), confirmed by stress and strain distribution across the device (Note S5 and Figures S16–S18, Supporting Information). The axial top side of the control FPSC without a NP design (C-FPSCs) experiences severe stress (436 MPa) and strain ($\approx 12\%$), whereas the NP-FPSC shows significantly lower stress (37 MPa) and strain ($\approx 2\%$). In the NP-FPSC, the maximum stress shifts from the functional layers to the PET protective layer, and the maximum strain shifts from the functional layers to the adhesive. This result demonstrates that the NP structure design effectively releases stress and strain on the functional layers of FPSCs.^[1a,8]

Motivated by the significantly reduced stress in NP-FPSCs, we evaluated their mechanical stability for devices fabricated with different adhesives. The devices underwent bending tests using a mechanical tester, following the bending test protocol for flexible photovoltaics.^[26] As the bending degree increased, higher stress was applied to the samples, with bending radius measured from the side view (Figure S19, Supporting Information). Notably, the bending tests were conducted with a bending radius of 4 mm , ensuring that the stress met the IEC 62715-6-3

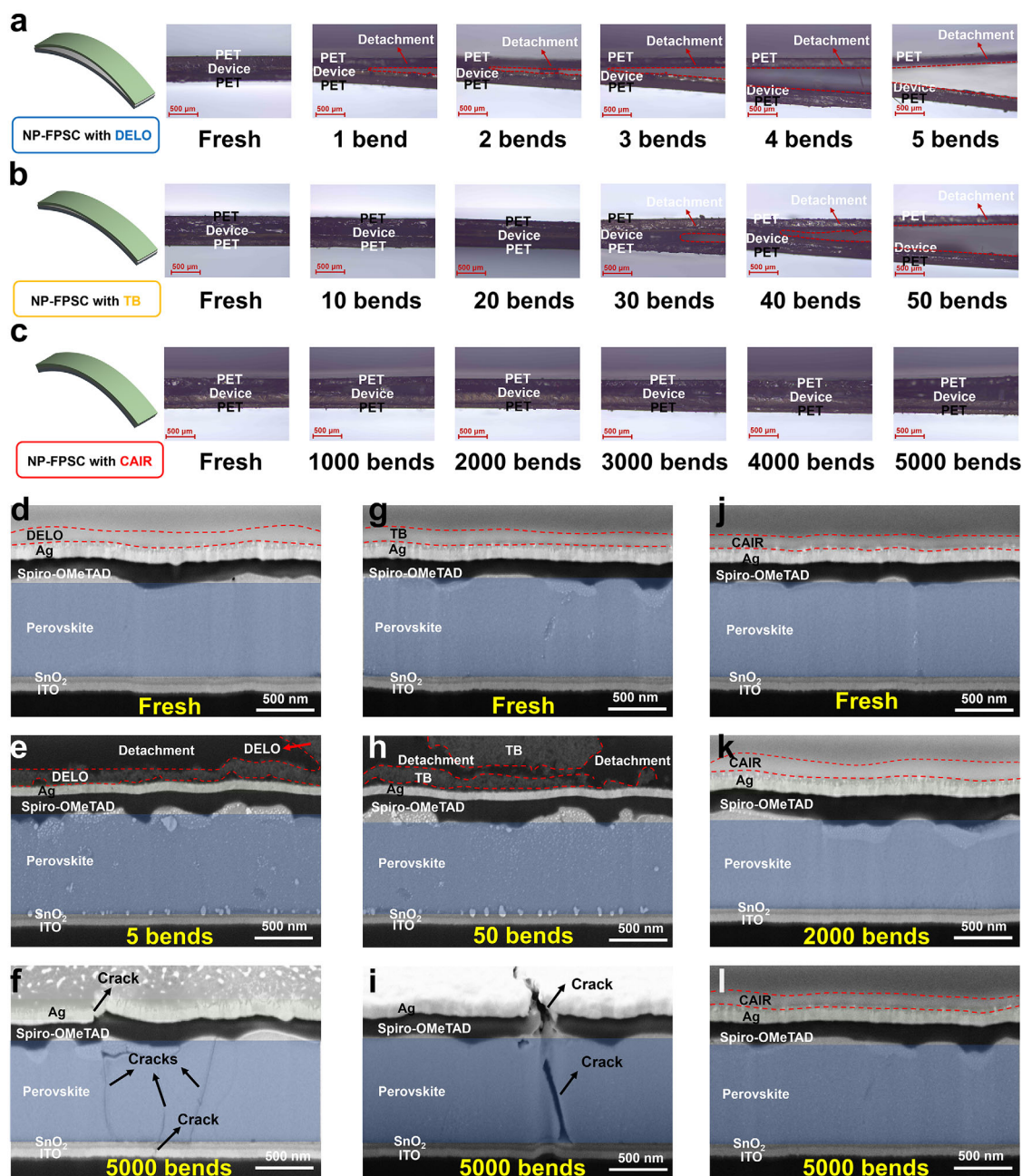


Figure 3. Cross-sectional optical microscope images of NP-FPSCs based on a) DELO, b) TB, and c) CAIR, before and after bending. Cross-sectional SEM images of NP-FPSCs based on DELO after bending for d) 0 bend, e) 5 bends, and f) 5000 bends; NP-FPSCs based on TB after bending for g) 0 bend, h) 50 bends, and i) 5000 bends; and NP-FPSCs based on CAIR after bending for j) 0 bend, k) 2000 bends, and l) 5000 bends, at a bending radius of 4 mm.

standards. We investigated whether NP-FPSCs with different adhesives showed detachment under continuous bending cycles. Cross-sectional optical microscope images revealed the structural evolution of the NP-FPSCs under continuous bending cycles at a radius of 4 mm (Figure 3a–c). The NP-FPSCs with DELO exhibited clear detachment between adhesives and devices after one bending cycle, with the PET protective layer completely separating from the devices within five cycles (Figure 3a). This behavior is attributed to the high Young's modulus of DELO generating high interlayer shear stress, whereas its low adhesion strength

failing to withstand such stress. In contrast, the TB adhesive with a lower Young's modulus can significantly delay detachment (Figure 3b), but the PET protective layer still completely detached from the device after 50 cycles, likely due to the ultralow adhesion strength of TB, which cannot resist interlayer shear stress, even when the stress has already minimized. Encouragingly, CAIR, exhibiting both an ultralow Young's modulus and high adhesion strength, showed significantly suppressed detachment behavior of NP-FPSCs, with the PET protective layer remaining firmly attached without cracks, even after 5000 bending cycles (Figure 3c).

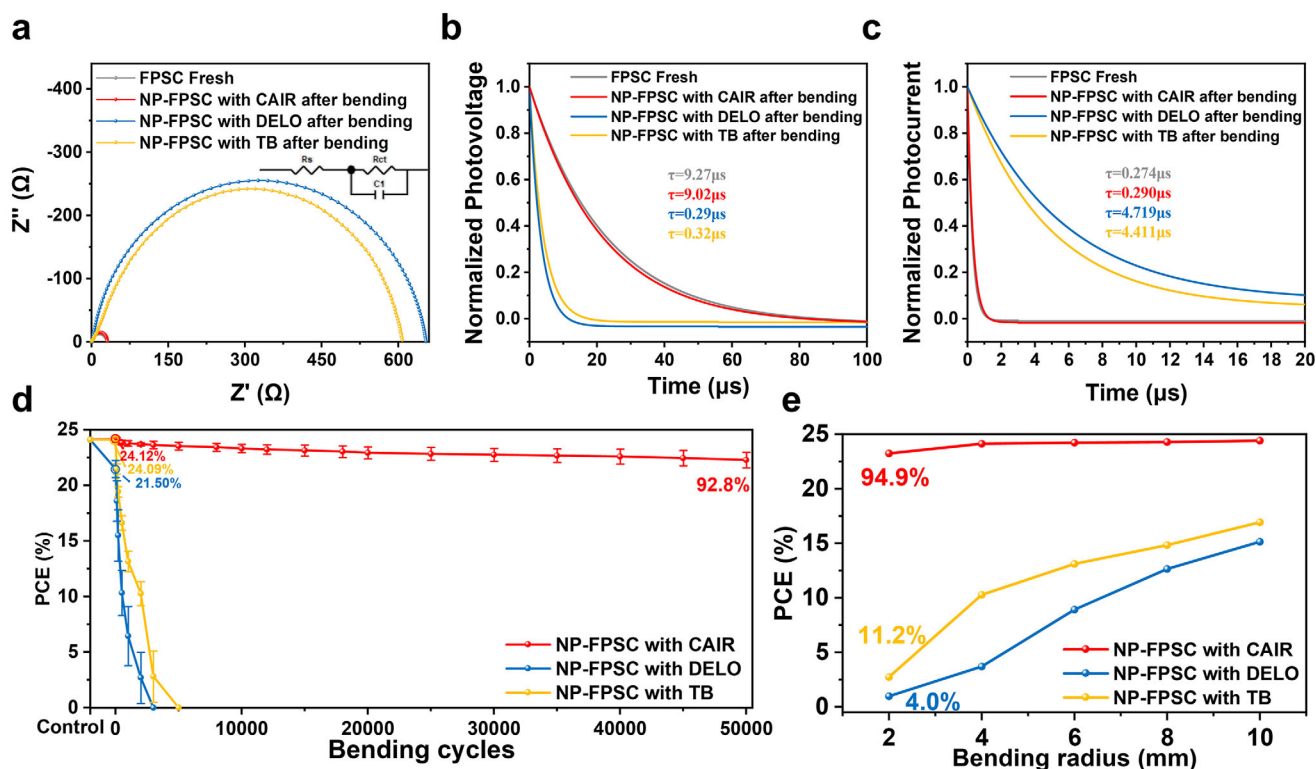


Figure 4. a) EIS Nyquist plots of Fresh FPSCs and NP-FPSCs with different adhesives after 5000 bending cycles at a radius of 4 mm (The plots obtained at a bias of 1.26 V in the dark). Inset: simulated equivalent circuit model. b) TPV results of Fresh FPSCs and NP-FPSCs with different adhesives after 5000 bending cycles at a radius of 4 mm. c) TPC results of Fresh FPSCs and NP-FPSCs with different adhesives after 5000 bending cycles at a radius of 4 mm. d) PCE decay of the NP-FPSCs with different adhesives at a radius of 4 mm (Control: before encapsulation). e) PCE decay of the NP-FPSCs with different adhesives versus bending radius after 2000 bending cycles.

To further investigate these behaviors, we examined the devices from a microscopic perspective using cross-sectional scanning electron microscopy (SEM) images of the entire flexible devices (Figure 3d–l). Significant detachment occurs at the interface between the DELO layer and Ag electrode (Figure 3e), with only a small amount of the adhesive remaining on the top of the Ag electrode after five bending cycles at a radius of 4 mm. This detachment disrupts the NP structure, subjecting all the functional layers to high stress and thus resulting in the formation of numerous cracks in all the layers (Ag electrode, Spiro-OMeTAD layer, perovskite layer, and ITO layer) after 5000 bending cycles (Figure 3f).^[5a,27] A similar evolution is observed in the NP-FPSCs with TB adhesive, where the PET protective layer detaches from the Ag layer after 50 bending cycles, and cracks appear in all the functional layers after 5000 bending cycles (Figure 3g–i). In contrast, the NP structure of the NP-FPSCs with CAIR adhesive is maintained, with the PET protective layer remaining tightly bonded even after 5000 bending cycles (Figure 3j–l). Notably, none of the functional layers in the NP-FPSCs based on CAIR adhesive exhibit fractures, even after 5000 bending cycles (Figure 3l). This result indicates that the exclusive encapsulation adhesive designed for NP-FPSC effectively mitigates structural failure by keeping the functional layers in NP.

The charge recombination dynamics and charge transport of the devices were explored to assess the effect of structural failure on performance. Nyquist plots derived from electrochemi-

cal impedance spectroscopy (EIS) measurements show that the NP-FPSC with CAIR exhibits almost unchanged series resistance (R_s ; 9.14 vs 11.50 Ω) and charge transfer resistance (R_{ct} ; 25.37 vs 29.72 Ω) after 5000 bending cycles at a radius of 4 mm (Figure 4a; Table S5, Supporting Information). In contrast, due to detachment-induced NP structure loss, the NP-FPSCs with DELO and TB exhibited significantly increased R_s (134.1 and 130.2 Ω , respectively) and R_{ct} (600.1 and 581.9 Ω , respectively), indicating higher series resistance and poorer interlayer electronic contact from structural failure. These results were corroborated by transient photovoltage (TPV) measurements (Figure 4b). After 5000 bending cycles, the NP-FPSC with DELO and TB showed photovoltage lifetimes of 0.29 and 0.32 μ s, respectively, which were shorter than those of the fresh FPSC (9.27 μ s) and bent NP-FPSC with CAIR (9.02 μ s). These behaviors demonstrate that structural failure blocks nonradiative recombination. In transient photocurrent (TPC) measurements (Figure 4c), the decay time of the NP-FPSC with CAIR was only negligibly changed (0.290 vs 0.274 μ s) after 5000 bends. However, the photocurrent lifetimes of the NP-FPSC with DELO and TB significantly increased to 4.719 and 4.411 μ s, respectively. This result confirmed that the NP design effectively suppressed bending-induced interlayer charge extraction. Then, the electrical properties of the transport layers were also carefully investigated. Notably, we employed a SnO_2 layer with a Young's modulus of ≈ 80 GPa as the electrode layer,^[28] and a spiro-OMeTAD layer with a Young's

modulus of ≈ 0.5 GPa as the hole transport layer.^[29] Due to the significantly lower Young's modulus of spiro-OMeTAD, this layer experiences negligible uniaxial tensile stress during bending, effectively preventing crack formation.^[5a] Consequently, our primary focus was on evaluating the electrical properties of the more mechanically sensitive SnO_2 layer. First, we investigated the electron mobility of the SnO_2 layer in a device structure consisting of PET/ITO/ $\text{SnO}_2/\text{C}_{60}/\text{BCP}/\text{Ag}$. For the NP-structured samples, the device was protected by a PET layer with different adhesives. As shown in Figure S20a (Supporting Information) and Table S6 (Supporting Information), the NP-structured sample with CAIR exhibited almost unchanged electron mobility (1.61×10^{-3} vs $1.59 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) after 5000 bending cycles at a radius of 4 mm. In contrast, due to detachment-induced loss of the NP structure, the NP-structured samples with DELO and TB exhibited a significant decrease in electron mobility (1.68×10^{-4} and $8.17 \times 10^{-4} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively). A similar trend was observed in the conductivity of the SnO_2 layer with the different structures. The NP-structured sample with CAIR maintained nearly constant conductivity, while the NP-structured samples with DELO and TB exhibited a significant reduction in conductivity (Figure S20b and Table S7, Supporting Information). These results strongly suggest that the NP structure effectively protects the flexible device from damage during multiple bending cycles.

Finally, we investigated the mechanism by which adhesives with different mechanical parameters influence the stability of NP-FPSCs. Notably, the PCE shows no decline for CAIR and TB adhesion, as evidenced by overlapping current density (J)–voltage (V) curves before and after encapsulation, whereas DELO adhesives cause significant PCE losses (Figure S21 and Table S8, Supporting Information). This decline may be attributed to the polar solvents in the DELO formulation.^[19,30] The PCE statistical distribution further confirmed the detrimental effects of the DELO adhesives on the photovoltaic performance of the device (Figure S22, Supporting Information). We also tracked the PCE evolution at the maximum power point (MPP) for the FPSCs with different adhesives to evaluate their impact on device stability. Figure S23 (Supporting Information) shows that the NP-FPSCs with CAIR and TB exhibited excellent operational stability with a T_{95} lifetime of >650 h, indicating that they do not adversely influence stability. In contrast, devices bonded with DELO exhibited severe efficiency degradation, with PCE dropping sharply to 42.4% within just 10 h.^[30,31] This sharply degradation may be attributed to interactions between the DELO adhesive and the Spiro-OMeTAD layer (Figure S24, Supporting Information). These results demonstrate that CAIR and TB are more stable than DELO when used between the Ag electrode and PET protective layer in NP-FPSCs. We further evaluated the mechanical stability of the NP-FPSCs. The PCE of the NP-FPSCs with TB rapidly decreased from 24.1% to 0% after 5000 bending cycles at a radius of 4 mm (Figure 4d) owing to severe detachment between TB and the Ag electrode disrupting the NP structure and causing structural failure. The NP-FPSCs with DELO experienced structural failure and additional damage to the perovskite crystals from polar solvents, with their efficiency decaying to 0% after only 3000 cycles. In contrast, the NP-FPSCs with CAIR retained 97.6% and 92.8% of their initial PCEs after 5000 and 50 000 bending cycles, respectively, meeting the IEC 62715-6-3 standards. To the best of our knowledge, this is the most mechanically stable FPSC

reported to date. The prolonged bending cycles are mainly attributed to the low maximum normal stress on the functional layers. Further, bending tests conducted at different radii (Figure 4e) showed that the NP-FPSC retained 94.9% of its initial PCE after 2000 bending cycles at a radius of 2 mm, thereby outperforming the NP-FPSCs with DELO (4.0%) and TB (11.2%). These results demonstrate that the undetached NP structure enhances the mechanical stability of FPSCs. Additionally, the NP structure also provides an encapsulation effect, greatly improving the long-time stability. As shown in Figure S25a (Supporting Information), NP-FPSCs with CAIR retained over 93% of their initial PCE after 500 h of storage in a humid atmosphere ($\approx 35 \pm 5\%$ RH), compared to the 58.2% PCE retention of C-FPSCs. We also evaluated the thermal stability of the devices stored in a nitrogen-filled glove box at 85°C. As shown in Figure S25b (Supporting Information), NP-FPSCs with CAIR maintained 84.2% of its initial PCE after aging for 500 h, whereas unencapsulated C-FPSCs exhibited a decline to 73.2%. The improved thermal stability of the encapsulated devices can be attributed to the barrier effect of the CAIR crosslinked network, which acts as a protective layer, effectively preventing the migration and diffusion of small molecules generated by thermal decomposition.^[32] These findings underscore that FPSCs with an NP structure exhibit exceptional mechanical, environmental, and thermal stability.

3. Conclusion

We identified that the primary cause of detachment between adhesives and electrodes in FPSCs is the high Young's modulus and low adhesion strength of commercial adhesives, which fail to endure interlayer shear stress during repeated bending cycles. To address this, we developed a crosslinkable polymer, AIR, with a linear polyisoprene backbone and acrylate side groups, enhancing flexibility and reducing chain segment movement. Under UV irradiation, AIR forms a robust, compact network with an ultralow Young's modulus and a high adhesion strength, ensuring a durable bond between the protective layer and FPSCs. The newly designed NP-FPSCs exhibit exceptional mechanical stability, retaining 92.8% of their initial efficiency after 50 000 bending cycles at a radius of 4 mm, surpassing the IEC 62715-6-3 standards. These advancements effectively improve FPSC mechanical stability and provide insights for developing other flexible optoelectronic devices, such as flexible sensors and organic light-emitting devices.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

T.Z. and X.P. contributed equally to this work. This work was supported by the National Natural Science Foundation of China (Grant Nos. 52325307 and 52273188), Department of Science and Technology of Jiangsu Province (No. BE2022023), the Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD), Collaborative Innovation Center of Suzhou Nano Science and Technology, and the Key Laboratory of Polymeric Materials Design and Synthesis for Biomedical Function, Soochow University. Suzhou Key Core Technologies Project (SYG2024146).

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

adhesive, flexible perovskite solar cells, mechanical stability, neutral plane

Received: January 24, 2025

Revised: May 8, 2025

Published online: June 10, 2025

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